A New Late Middle Cambrian Paleomagnetic Pole for the Ellsworth Mountains, Antarctica

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ABSTRACT

A paleomagnetic study of the late Middle to possibly early Late Cambrian Liberty Hills Formation in the Ellsworth Mountains, Antarctica, reveals a stable magnetization with positive fold and reversal tests. The paleopole is based on 16 sites from volcanic and sedimentary rocks and lies at lat 7.3°N and long 326.3°E (A_{95} = 6.0°). The new paleomagnetic data support the view that the Ellsworth Mountains are part of a microplate—the Ellsworth-Whitmore Mountains crustal block—that rotated independently of the main Gondwana continental blocks during breakup. The Liberty Hills pole differs from both previous poles recovered from Cambrian rocks in the Ellsworth Mountains and from the available Gondwana reference pole data. Our pole indicates a more northerly prebreakup position for the Ellsworth Mountains than previously suggested, contradicting the overwhelming geologic evidence for a prebreakup position close to southern Africa. The reasons for this are uncertain, but we suggest that problems with the Gondwana apparent polar wander path may be important. More well constrained, early Paleozoic paleomagnetic data are required from the Ellsworth Mountains and the Gondwana continents if the data are to constrain further the Middle-Late Cambrian location of the Ellsworth-Whitmore Mountains block.

Introduction

The Ellsworth Mountains are part of the allochthonous Ellsworth-Whitmore Mountains (EWM) crustal block (fig. 1*A*), one of five forming West Antarctica (Dalziel and Elliot 1982; Storey et al. 1988; Maslanyj and Storey 1990). The prebreakup positions of these terranes are crucial to Gondwana reconstructions and breakup models. Geologic comparisons of the early Paleozoic strata of the EWM and other sectors of the paleo-Pacific margin of Gondwana allow the general prebreakup position of the EWM block to be located with some confidence along the Gondwana margin. The East Antarctic sector of the Gondwana margin was characterized by an early Paleozoic orogenic event, the Ross orogeny. In the Ellsworth Mountains, despite recent arguments in favor of a Ross-age (End Cambrian) orogenic event (Duebendorfer and Rees 1998), the weight of structural, stratigraphic, and geochemical evidence strongly supports the absence of contractional/orogenic deformation of the Cambrian succession (Webers et al. 1992*b*; Goldstrand et al. 1994; Curtis 1998*b*; Curtis and Lomas 1999; Curtis et al. 1999). Furthermore, the application of high-precision single-zircon dating techniques on pre–Table Mountain Group Sandstone (pre-Ordovician) has led to the identification of Middle Cambrian continental rift deposits in South Africa (Barnett et al. 1997; Armstrong et al. 1998), which form an excellent correlative succession to a similar rift-related volcanic succession in the EWM (Curtis et al. 1999). This recently recognized correlation very strongly supports early Paleozoic Gondwana reconstructions that place the EWM adjacent to, or within, the Natal Embayment before the supercontinent breakup during the Middle Jurassic (fig. 1*B*; e.g., Schopf 1969; Dalziel and Grunow 1992; Curtis and Storey 1996; Dalziel 1997). Such a position juxtaposes the only two geologic provinces along the Gondwana paleo-Pacific margin that preserve virtually complete Paleozoic successions containing evidence for Middle to Late Cambrian continental rifting, as opposed to orogenesis (Curtis et al. 1999). In addition, this position places the EWM within a belt of Grenvillian-

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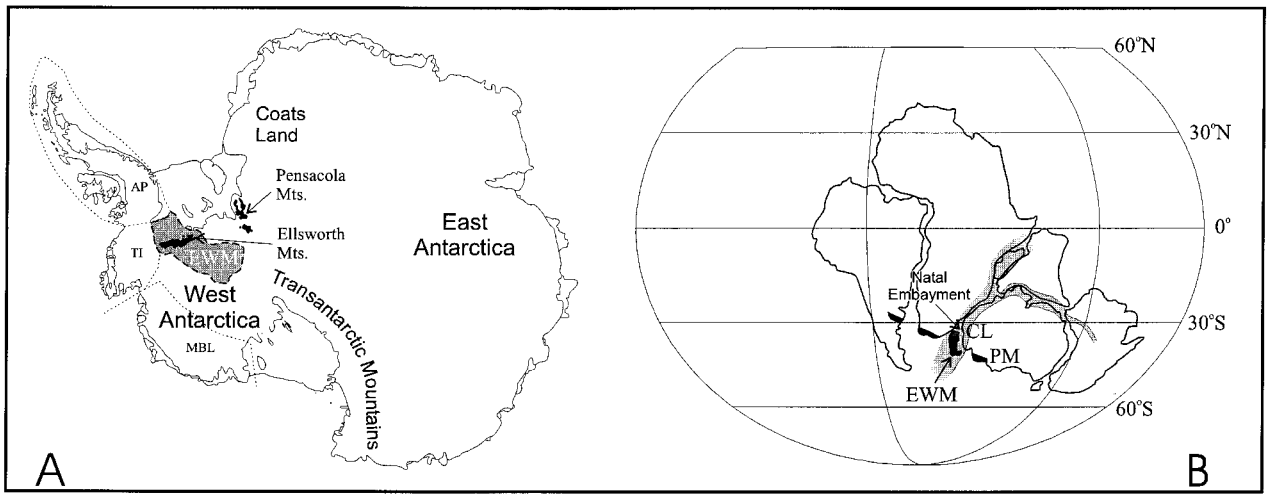


Figure 1. *A,* Position of the Ellsworth Mountains within the Ellsworth-Whitmore Mountains (*EWM*) block and position of the EWM within Antarctica. For the purposes of this article, the EWM includes the small adjoining Haag Nunataks block. Positions of the other West Antarctic continental terranes are also shown: *AP* = Antarctic Peninsula; *TI* = Thurston Island; *MBL* = Marie Byrd Land; modified from Storey et al. (1988). *B,* A prebreakup position for the EWM, within the Natal Embayment between southern Africa and Coats Land (redrawn from Dalziel and Grunow 1992). Shaded area shows extent of Grenvillian basement (Moores 1991; Storey et al. 1994). Black area shows the structural trend of the Permo-Triassic Gondwanan orogeny. *CL* = Coats Land; *PM* = Pensacola Mountains.

age (1–1.2 Ga) crust (Grantham et al. 1997), which satisfies isotopic evidence that indicates the EWM is underlain by crust with a mean age of 1123 Ma (Storey et al. 1994; Curtis et al. 1999). Restoring the EWM to a prebreakup position in the Natal Embayment also aligns the structural trend of the Permo-Triassic Gondwanan orogeny (fig. 1*B*).

Three paleomagnetic poles are currently available to help constrain the pre- and synbreakup position of the EWM. Watts and Bramall (1981), using data from the Late Cambrian sedimentary rocks of the Frazier Ridge Formation in the Ellsworth Mountains, suggested an approximate 80° counterclockwise rotation of the EWM. This pole position and the tectonic interpretation are supported by data from 12 samples analyzed from undated sedimentary rocks in the Nash Hills, some 250 km south of the Ellsworth Mountains themselves (Grunow et al. 1987). Grunow et al. (1987), based on data from Middle Jurassic granitoids, suggested a 15° counterclockwise rotation of the EWM, postdating the acquisition of the primary remanence. Together, the three poles have been interpreted as supporting a prebreakup position close to southern Africa.

This article presents a new, well-defined, and dated pole from the late Middle to possibly early Late Cambrian Liberty Hills Formation in the southern Ellsworth Mountains. Using this pole in conjunction with geologic evidence, we assess the validity of the available paleomagnetic data and the reconstruction models for the EWM within Gondwana.

Regional Geology and the Liberty Hills Formation

The Ellsworth Mountains expose a 13-km-thick Lower Cambrian through Permian stratigraphic succession (Webers et al. 1992*b*). The southern half of the Ellsworth Mountains, the Heritage Range (fig. 2), is predominantly composed of Cambrian-age sedimentary and volcanic rocks of the Heritage Group, of which the laterally equivalent Frazier Ridge, Springer Peak, and Liberty Hills Formations (Webers et al. 1992*a*) are the most extensively exposed. These formations are principally composed of argillite, greywacke, and conglomerate, with lesser amounts of basic volcanic and volcaniclastic rock occurring in the Springer Peak and Liberty Hills Formations. The geochemistry of these late Middle Cambrian volcanic rocks is consistent with their eruption in a continental-rift setting (Curtis et al. 1999). Overlying these formations, along a sequence-bounding unconformity, are the laterally discontinuous, northward-thinning limestones of the Upper Cambrian Minaret Formation. The Minaret Formation is, in turn, overlain by a thin se-

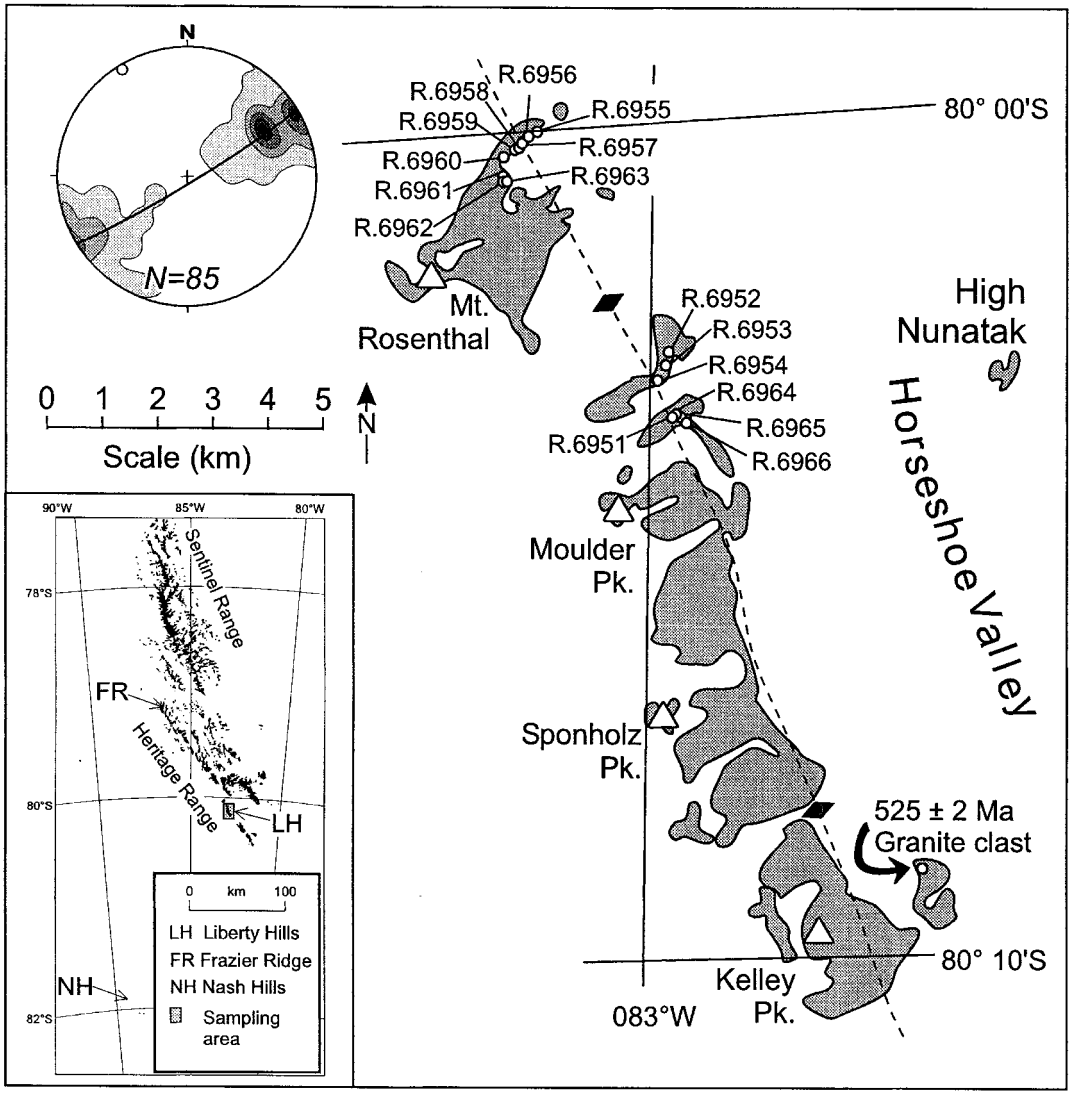


Figure 2. Map of Liberty Hills showing paleomagnetic sampling sites and collection site of dated granite clast. Dashed line shows axial trace of the Liberty Hills anticline. A contoured stereoplot of poles to bedding reveals the subhorizontal nature of the fold axis. *Inset*: Location of Liberty Hills within the Ellsworth Mountains and the location of the sampling sites of Watts and Bramall (1981) in the Frazier Ridge Formation and Grunow et al. (1987) in the Nash Hills.

quence of siltstones and shales that comprise the transition beds (Spörli 1992). Stratigraphically overlying the transition beds are the quartzites of the basal Crashsite Group (Spörli 1992), interpreted as representing the establishment of a passive margin during the Early-Middle Ordovician (Curtis and Lomas 1999; Curtis et al. 1999). Despite localized unconformities along the base of the Crashsite Group (Goldstrand et al. 1994; Duebendorfer and Rees 1998), the general conformity along, and structural continuity across, this contact defines it as a sequence-bounding unconformity (Curtis and Lomas 1999).

The Liberty Hills Formation is best exposed within the Liberty Hills of Horseshoe Valley, southern Heritage Range (fig. 2). The formation is characterized by a succession of conglomerate, quartzite, argillite, and mafic lava that fines upward and to the northeast (Webers et al. 1992*a*). The basic volcanic rocks have a maximum thickness of 750 m at Moulder Peak and rapidly thin northward until they are absent at Mount Rosenthal. The rocks

of the Liberty Hills display evidence of only a single, intense deformation event (Permo-Triassic Gondwanan orogeny), which produced a first-order, asymmetric anticline that possesses a subvertical to overturned northeastern limb (Curtis and Lomas 1999) and a near horizontal fold axis that runs through the eastern side of the hills (dashed line in fig. 2). Second-order parasitic folds are also present and observed at both Mount Rosenthal and Moulder Peak.

Age Constraints for the Liberty Hills Formation. Recent studies have provided several new radiometric ages for the Cambrian, allowing much tighter constraints to be placed on the geologic time scale (Davidek et al. 1998; Landing et al. 1998). For the purposes of this article, the end of the Early Cambrian is taken to be 510 Ma and the Late Cambrian-Ordovician boundary is taken to be 490 Ma. The absence of radiometric ages makes defining an upper limit for the Middle Cambrian somewhat problematic, and the 500-Ma age suggested by Shergold (1995) has been used here.

A minimum age for the Liberty Hills Formation is provided by the overlying Minaret Formation, biostratigraphically dated as early Late Cambrian (e.g., Webers et al. 1992*c*) but possibly as old as latest Middle Cambrian (Jago and Webers 1992). This age equates to a minimum age of approximately 500 Ma.

A maximum age of 525 ± 2 Ma is provided by zircons from a granite clast collected from a conglomerate unit to the east of Kelley Peak (fig. 2). Zircons were separated for U-Pb analysis using standard techniques. Grains selected for analysis were prismatic, between 100 and 200 μm in length, with no visible cracks or inclusions. All fractions were abraded (Krogh 1982). Single grains or small fractions (up to six grains) were processed using the methods of Krogh (1973) and Corfu and Ayres (1984), using a mixed $^{205}Pb + {}^{235}U$ isotopic tracer (Krogh and Davis 1985). The data are given in table 1; data reduction was carried out using the methods of Ludwig (1989, 1990). Uncertainty in the age is reported at 95% confidence limits. Procedural blanks were 1.5 pg for Pb and 0.5 pg for U. Three strongly abraded zircon fractions lie on, or close to, concordia, whereas five less strongly abraded fractions are discordant, with $^{206}Pb/^{238}U$ ages between 465 and 480 Ma (fig. 3). Seven of the eight analyzed zircon fractions plot on a discordia line, indicating an upper intercept age of 525 ± 2 Ma, with modern-day Pb loss. If one allows time for unroofing, erosion, and deposition of the granite clast, over a relatively modest time period of 10 m.yr., a depositional age range of ~515–500 Ma is suggested for the Liberty Hills Formation. Given that the Liberty Hills Formation is laterally equivalent to the biostratigraphically well dated late Middle Cambrian Springer Peak Formation (Jago and Webers 1992), the actual age of the Liberty Hills Formation is more likely to be in the range of 505–500 Ma.

Paleomagnetic Analysis

During the 1996–1997 austral summer, samples were collected from 16 sites for paleomagnetic analysis. Seven sites were located in basic volcanic rocks on the eastern flank of Moulder Peak (R.6951–R.6954 and R.6964–R.6966) and nine were located in fine sandstones and argillites on the northeast ridge of Mount Rosenthal (fig. 2). Sites at Mount Rosenthal were collected from both limbs of the Liberty Hills anticline, while the Moulder Peak sites were taken from the eastern limb only. However, the second-order parasitic folding means that sites with opposing tilt directions are sampled within the same limb of the Liberty Hills anticline; this therefore allows a fold test to be performed on both the sedimentary and volcanic rocks. Natural remanent magnetizations (NRM) were measured using a two-axis CCL cryogenic magnetometer at the Oxford University paleomagnetic laboratory. Specimens were stepwise demagnetized using two-axis tumbling alternating field or thermal methods through at least 12 steps until they became unstable, or to 100 mT or 700°C, respectively. Thermal demagnetization was more efficient at removing secondary magnetic components and was, therefore, the more commonly employed method. Bulk susceptibility was measured after each thermal demagnetization step to detect heat-induced mineralogical changes.

Demagnetization behavior was variable, and most specimens display multicomponent NRMs. Where possible, remanence directions were determined using principal component analysis (Kirschvink 1980) applied to colinear segments converging toward the origin. Components were considered stable when they were defined by at least three points on vector endpoint diagrams and had a maximum angular deviation not exceeding 15°. The overlapping nature of the blocking spectra meant that in some cases it was impossible to define linear components, and remagnetization circles were often required to recover the direction (McFadden and McElhinny 1988). Sample means were combined to site means, and site means, to a formation mean using standard Fisher (1953) statistics.

Magnetic Carriers. The nature of the magnetic carriers was investigated using acquisition of iso-

Table 1. Isotope Data for the Granite Clast Sample from Kelley Peak

	Fraction[a]	Weight[b] (μg)	U (ppm)	Pb (ppm)	Common Pb (pg)	$^{206}Pb/^{204}Pb$[c]	$^{208}Pb/^{206}Pb$[d]	$^{206}Pb/^{238}Pb$[d]	$^{207}Pb/^{235}Pb$[d]	$^{207}Pb/^{206}Pb$[d]	$^{207}Pb/^{206}Pb$[d] Age (Ma)	ρ[e]
Granite clast, Liberty Hills Formation, sample R.6683.3A:												
1A	2 grains, 100 μm	22.8	272.5	23.8	6.5	6536	.1481	.08361 ± .21	.6668 ± .23	.05784 ± .10	524 ± 2	.91
1B	1 grain, 200 μm	10.8	241.3	19.5	5.9	2873	.1428	.07675 ± .36	.6134 ± .47	.05796 ± .29	528 ± 6	.79
1C	4 grains, 150 μm	20.5	286.7	22.4	4.7	8577	.1466	.07486 ± .23	.5978 ± .27	.05792 ± .14	527 ± 3	.86
1D	5 tips, 100 μm	26.1	445.1	34.4	8.7	7518	.1381	.07446 ± .23	.5939 ± .24	.05785 ± .08	524 ± 2	.94
Granite clast, Liberty Hills Formation, sample R.6683.3B:												
1A	1 grain, 300 μm	43.6	201.8	16.2	4.9	12,348	.1682	.07588 ± .24	.6046 ± .26	.05778 ± .10	522 ± 2	.92
1B	6 grains, 100 μm	45.5	233.1	20.4	6.2	11,937	.1462	.08398 ± .23	.6707 ± .25	.05793 ± .09	527 ± 2	.94
1C	5 grains, 100 μm	34.8	209.9	19.4	34.5	1191	.1468	.08490 ± .22	.6771 ± .25	.05785 ± .12	524 ± 3	.87
1D	?1–2 grains, 60 μm	4.7	656.6	51.5	2.8	10,804	.1459	.07522 ± .29	.6096 ± .38	.05878 ± .25	559 ± 5	.76

[a] All zircon grains were picked from the nonmagnetic/nondiamagnetic split on a Frantz LB-1 Magnetic Barrier Separator. All grains were abraded before dissolution.
[b] Errors on sample weights, and therefore on U and Pb concentrations, are approximately 20%.
[c] Measured ratio corrected for fractionation and spike.
[d] Corrected for fractionation, spike, laboratory Pb and U blanks, and initial common Pb (Stacey and Kramers 1975). Assumed Pb blank composition is $^{206}Pb/^{204}Pb = 18.30$; $^{207}Pb/^{204}Pb = 15.56$; $^{208}Pb/^{204}Pb = 37.63$. Assumed blanks are 1.5 pg for Pb and 0.5 pg for U. Errors are quoted as percentages at the 2σ level. Data were reduced using the method of Ludwig (1989). Errors on measured ages propagated through the data reduction calculations were ±2 standard errors of the mean.
[e] Correlation coefficient of $^{207}Pb/^{235}U$ to $^{206}Pb/^{238}U$ is calculated using the procedures and algorithm of Ludwig (1989).

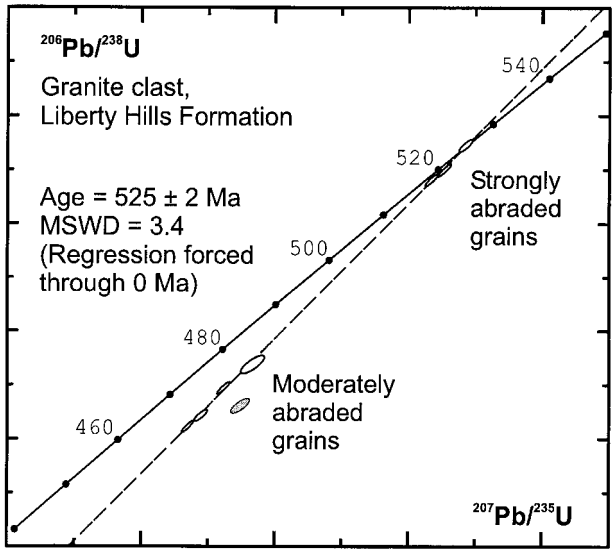


Figure 3. Concordia diagram for granite clast from Kelley Peak (see fig. 2). Seven zircon fractions and single grains plot on a discordia indicating an age of 525 ± 2 Ma.

thermal remanent magnetization (IRM) experiments and subsequent demagnetization of three orthogonal IRMs (Lowrie 1990). These experiments suggest a range of magnetic carriers. For the sedimentary rocks from Mount Rosenthal, saturation magnetization was achieved in an applied field of approximately 300 mT, and the dominant low coercivity IRM was demagnetized steadily by thermal treatment up to 580°C (fig. 4*A*). Together the data indicate that magnetite and possibly some titanomagnetite carry the induced remanence. Specimens from the Moulder Peak volcanic rocks contain a low coercivity fraction, indicated by a decreased slope of the IRM acquisition curve at 100–300 mT and a high-coercivity fraction that remains unsaturated at the peak applied field of 800 mT (fig. 4*B*, 4*C*). Thermal demagnetization data of the IRM are consistent with magnetite and hematite in the samples in varying proportions. The hematite in the volcanic rocks results from deuteric alteration of the lavas that can be seen to predate the deformation (Bauer 1992). The alteration is probably a consequence of early volcanic fluids and, therefore, occurred rapidly after extrusion. A primary remanence carried by hematite is therefore likely to have been acquired soon after deposition.

High-temperature susceptibility experiments on powdered samples, carried out using a CS2 furnace attached to a KLY-2 bridge, yielded curie temperatures consistent with the mineral identifications made in the IRM experiments. However, they also revealed that some samples undergo alteration on heating. This is characterized by the growth of magnetite in the volcanic rocks and titanomagnetite (unblocking temperature ~450°C) in the sedimentary rocks. Bulk susceptibility data obtained during thermal demagnetization demonstrate that the alteration is more pronounced in the temperature-dependent susceptibility studies, almost certainly a consequence of working with powdered samples. Furthermore, the bulk susceptibility measurements show that the majority of the alteration occurs above the demagnetization range at which the characteristic remanent magnetization (ChRM) is isolated. Checks were made on the raw demagnetization data, and no uniform direction consistent with a laboratory-induced remanence was recognized. The alteration does not therefore affect the ChRM.

Results. Examples of demagnetization behavior are shown in figure 5. The examples chosen demonstrate the extremes of the observed behavior, particularly in terms of unblocking temperatures. The sedimentary rocks collected from Mount Rosenthal are weakly magnetized and record NRM intensities in the range 10^{-4}–10^{-2} Am^{-1}. All specimens show similar demagnetization behavior, characterized by multiple magnetic components with overlapping unblocking spectra. Samples fully demagnetize by approximately 520°–580°C, indicating that magnetite, with possibly some titanomagnetite, carries the NRM, consistent with the magnetic carriers identified from the IRM and high-temperature susceptibility studies. Neither the rock magnetic tests nor the analysis of thin sections reveals any hematite in the sedimentary rocks. The lowest unblocking temperature component is recovered below 100°C and has sample directions scattered within a site. These are interpreted as a short-term viscous remanent magnetization (VRM) acquired after sampling. An intermediate component, held between 100°C and 250°–300°C, succeeds the low unblocking temperature component. In most cases, all samples in a site possess a consistent intermediate direction, and although there are variations in the declination, all are steeply inclined in situ. These are interpreted as a present-field overprint acquired as a long-term VRM. A few sites have samples with more scattered intermediate-unblocking-temperature components, which may represent shorter-term VRMs. With continued demagnetization to higher temperatures, the directions move away from these intermediate groups until the sample magnetization becomes unstable at 500°–580°C. In a very few samples, stable endpoints could be

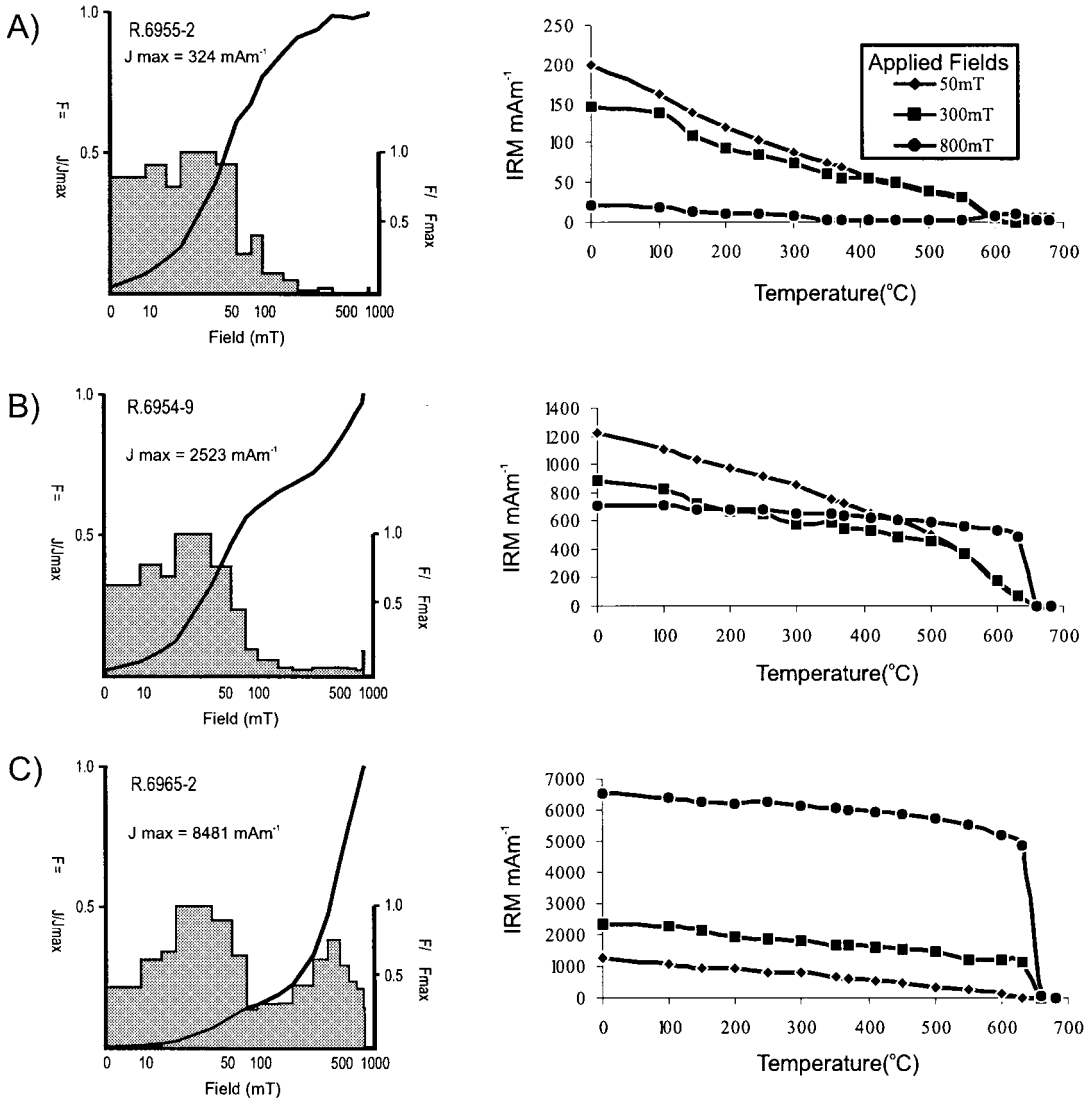


Figure 4. Isothermal remanent magnetization (IRM) acquisition and subsequent demagnetization of three orthogonally applied IRMs. *A,* Mixed titanomagnetite/magnetite carriers in a sedimentary rock from Mount Rosenthal. *B,* Mixed magnetite/hematite carriers from a volcanic rock at Moulder Peak. *C,* Hematite dominant magnetite/hematite mixed carriers from a volcanic site at Moulder Peak.

isolated, defined by a few points. The high-temperature data for most samples, though, when viewed on Zijderveld plots, bypass the origin, and on stereoplots, are strung along great-circle paths. This indicates that a high-temperature component is present but its direction could not be isolated (fig. 5*A*–5*C*). In this case, remagnetization circles were used to recover the highest-temperature component and, subsequently, the ChRM for the site (fig. 6*A*–6*C*).

Irrespective of the direction of the intermediate component, all sites have a ChRM directed to the northeast or southwest with a shallow inclination in tilt-corrected coordinates. One of the sites has a relatively large confidence limit (site R.6957, A_{95} = 16°); however, the site mean has a good precision parameter and accords with the rest of the data. It is, therefore, included with the rest of the data for mean calculations. The magnetic polarity recorded by the samples is consistent within the individual sites sampled in the sedimentary rocks, but normal and reverse polarities are recorded at different site localities (table 2).

Volcanic rocks from the Moulder Peak locality show three types of demagnetization behavior, each characterized by the magnetic remanence carriers.

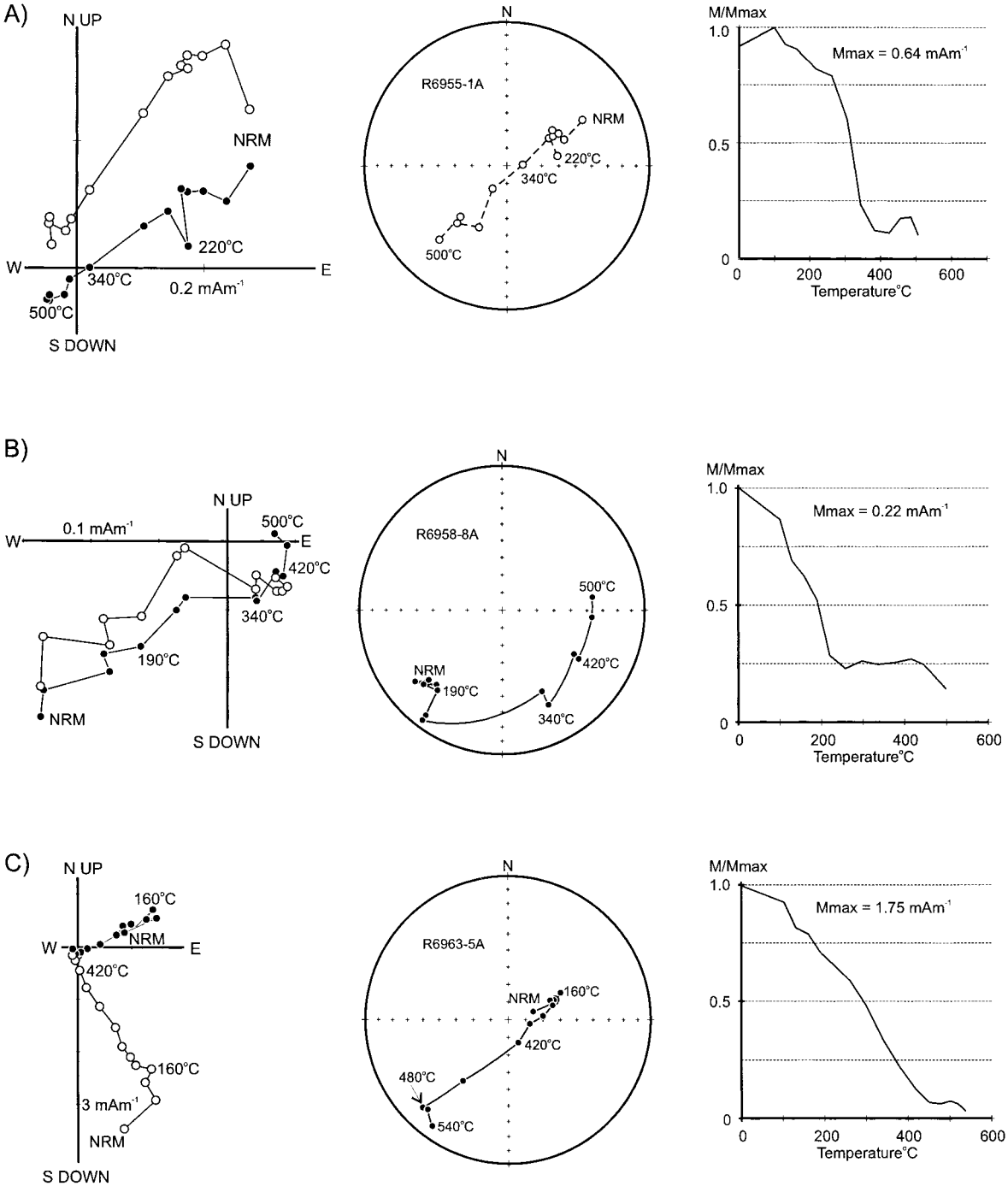


Figure 5*A*–5*C*

Figure 5. Examples of the various demagnetization behavior. On orthogonal plots, open symbols are vertical projections; solid symbols are horizontal projections. On equal-area stereoplots, open symbols are upper hemisphere projections; solid symbols are lower hemisphere projections. Unblocking temperatures are shown by normalized intensity versus demagnetization temperature. Plots *A*, *B*, and *C* show examples from sedimentary rocks from Mount Rosenthal from east limb, core, and west limb of the fold, respectively. Plots *D*, *E*, and *F* show examples from volcanic rocks with titanomagnetite/magnetite, titanomagnetite/magnetite with dominant or finer-grained magnetite, and hematite-dominated remanence carriers, respectively. All plots show data after tilt correction.

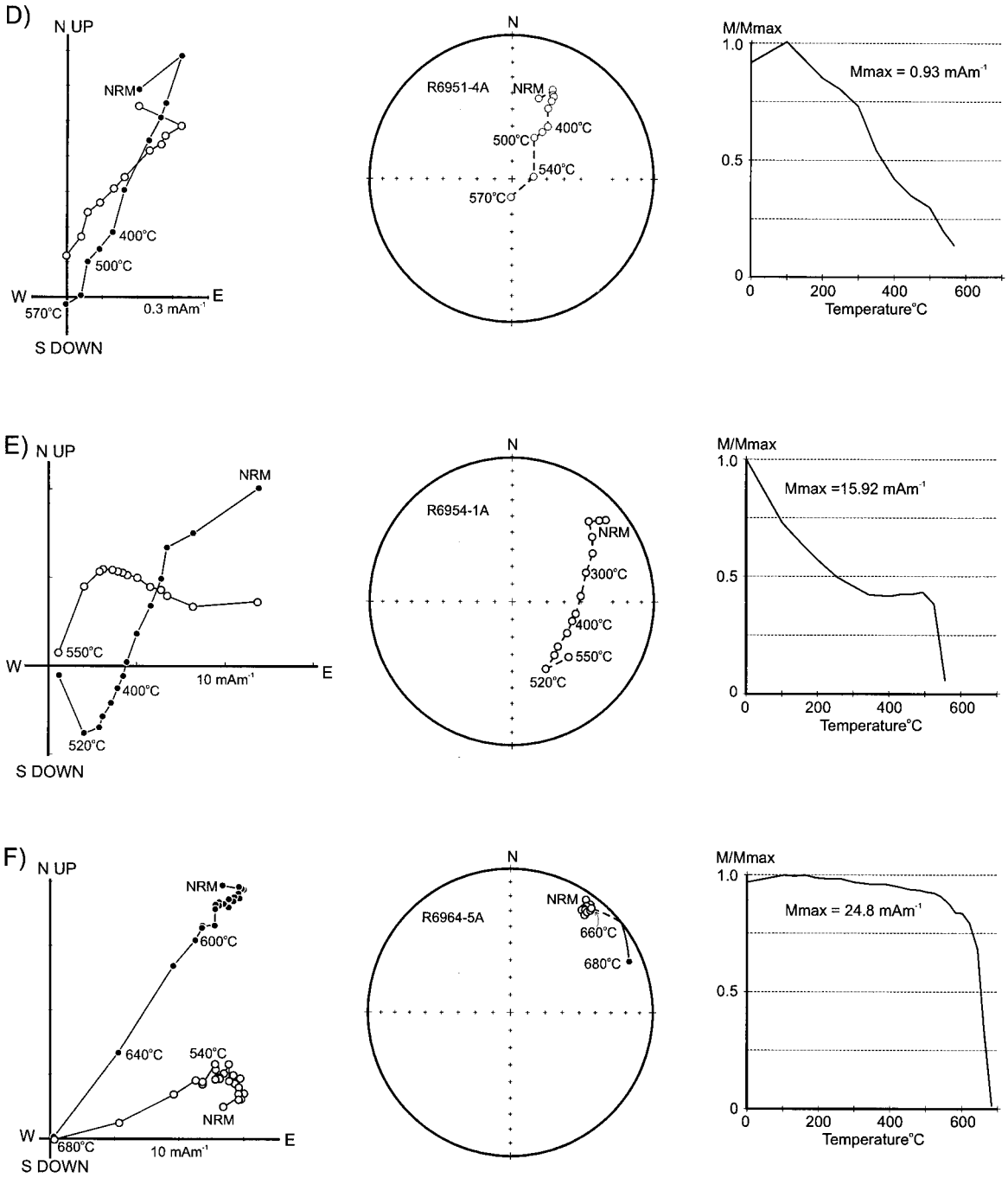


Figure 5*D–5F*

Type I samples, observed from sites R.6951 and R.6966, are weakly magnetized, with NRMs typically $<10^{-3}$ Am^{-1}. Thin-section analysis and IRM studies reveal the presence of some minor hematite, localized around rare, altered, ferromagnesian minerals. However, the hematite does not contribute to the NRM, and the samples display a (titano) magnetite carrier of the natural magnetization. The style of demagnetization is similar to that observed in the sedimentary rocks: a site-inconsistent, low-unblocking-temperature component; a steeply inclined intermediate component removed by ~300°C; and a higher-temperature component that remains stable up to ~580°C. As with the sedi-

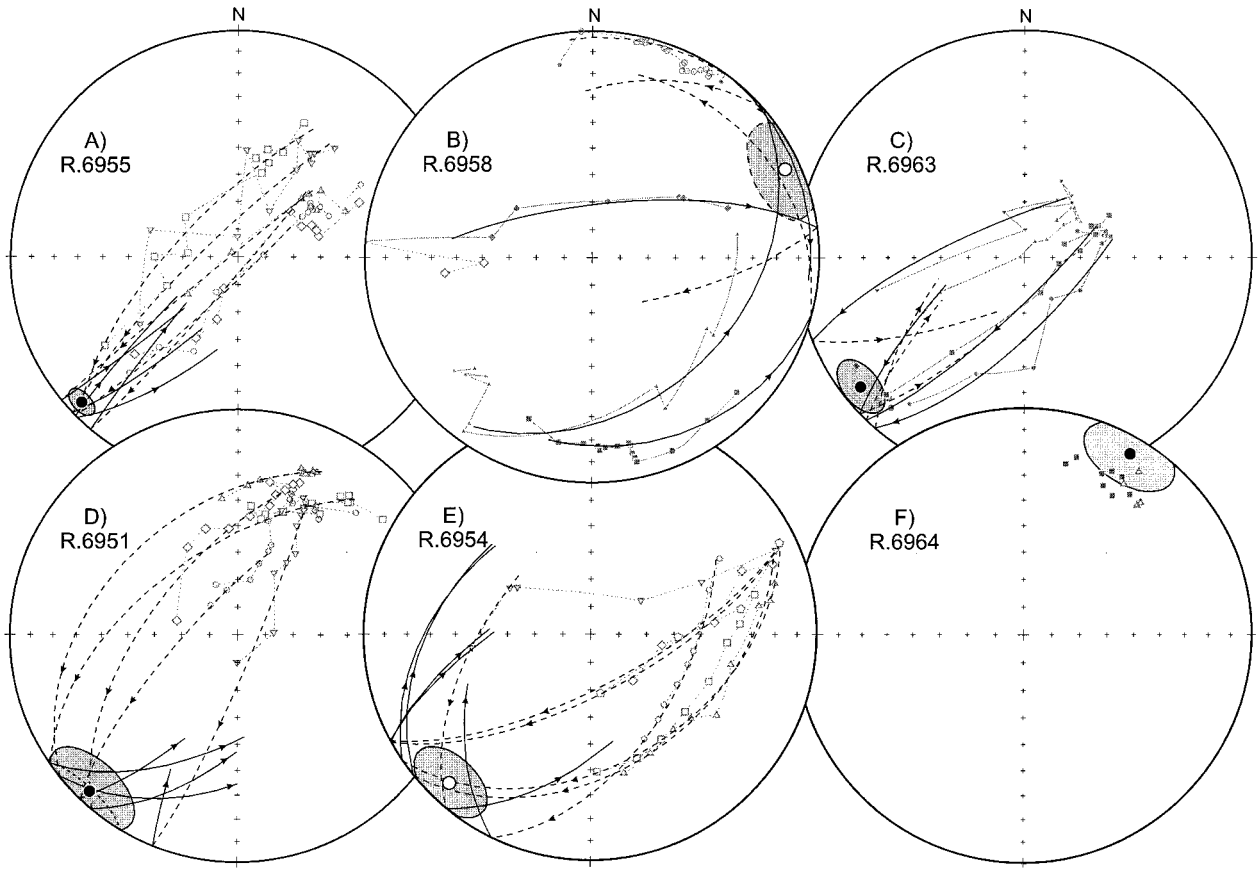


Figure 6. Equal-area stereoplot projections showing how site mean directions were obtained from intersecting great circles or stable endpoints. Open symbols are upper hemisphere projections; solid symbols are lower hemisphere projections. Plots *A–C* show examples of sedimentary rocks from Mount Rosenthal. Plots *D–F* show examples of volcanic rocks from Moulder Peak. For clarity, some low unblocking temperature demagnetization steps have been removed, and only a selection of the great-circle paths that were actually used to obtain the mean are shown in plots *A–E*. In plot *F*, stable endpoints that have been inverted to obtain the mean are shown as triangles. All plots show data after tilt correction.

mentary rocks, the higher-temperature component bypasses the origin on Zijderveld plots and tracks out along a great-circle path on a stereoplot. For both of these sites, intersecting remagnetization circles were used to obtain the ChRM and the site mean direction (fig. 6*D*).

Type II demagnetization, seen in sites R.6953 and R.6954, has samples with higher NRM intensities than before, up to 10^{-2} Am^{-1}. Magnetite and hematite are identified in these samples from thin sections and from IRM and high-temperature susceptibility studies (fig. 4*B*). However, demagnetization of the NRM reveals that the hematite does not contribute to the natural remanence, and all samples demagnetize completely by 580°C (fig. 5*E*). For Type I samples, high-temperature demagnetization steps are strung out along a great-circle path from (in this case) a low-temperature component oriented randomly within a site (fig. 5*E*). When viewed on Zijderveld plots, some samples appear to have a linear, origin-bound, high-temperature component (e.g., fig. 5*E*). However, these high-temperature components are not present in all samples; when they are present, they are typically based on only a single data point and are directionally inconsistent within and between the two sites. In contrast, great circles fitted through the lower unblocking temperature data give good intersections and site mean directions consistent with data from the other sites (fig. 6*E*). The major difference between the Type I and Type II behavior is the greater proportion of pure, or finer-grained, magnetite in the Type II samples. This leads to distinct drop in intensity at 550°–580°C, rather than the steady de-

Table 2. Paleomagnetic Data from Liberty Hills

Site	Lithology	Location		n/N[a]	In situ[b]		Tilt correction[c]	Tilt corrected[b]				VGP[d]		K	A_{95} (°)
		Latitude (°S)	Longitude (°W)		Dec	Inc		Dec	Inc	k	α_{95} (°)	Latitude (°N)	Longitude (°E)		
R.6951	Porphyritic basalt	80.07	82.95	12/13	233.8	76.0	130/70	223.3	6.3	11.3	13.9	10.3	321.2		
R.6952	Reddened basalt	80.04	82.98	7/9	244.5	−63.5	322/87	238.0	22.8	30.0	13.6	17.0	337.2		
R.6953	Porphyritic basalt	80.04	82.98	8/9	230.3	−68.2	322/87	231.3	18.8	363.6	3.4	15.8	330.1		
R.6954	Basaltic andesite	80.04	82.98	11/13	88.7	−79.3	320/88	223.2	−10.3	16.4	12.1	2.1	320.0		
R.6955	Argillite	80.00	83.17	9/10	138.9	−62.8	342/109	226.9	6.9	176.0	4.4	10.2	324.6		
R.6956	Argillite	80.00	83.17	6/7	339.5	74.7	341/86	55.7	4.3	85.0	9.5	3.5	332.6		
R.6957	Fine-grained sandstone	80.00	83.17	6/6	233.7	78.2	152/84	240.3	−5.7	30.3	16.0	2.1	337.1		
R.6958	Fine-grained sandstone	80.00	83.17	8/8	258.2	−81.6	157/90	65.4	−8.2	21.7	14.1	8.2	343.2		
R.6959	Argillite	80.01	83.18	7/9	109.5	−87.4	322/88	229.8	−3.4	83.7	8.1	4.7	326.8		
R.6960	Argillite	80.01	83.18	6/7	223.1	41.4	152/61	227.2	−17.5	54.4	11.9	−2.1	323.3		
R.6961	Argillite	80.02	83.19	10/11	52.9	−46.4	133/49	49.8	2.2	17.7	14.2	5.3	326.9		
R.6962	Argillite	80.02	83.19	10/10	234.7	74.2	147/42	236.3	32.2	16.0	13.7	22.8	336.2		
R.6963	Argillite	80.02	83.19	9/10	225.7	53.3	150/45	231.4	9.1	43.4	8.5	10.8	329.3		
R.6964	Porphyritic basalt	80.07	82.95	12/13	12.6	−41.6	160/70	30.4	8.7	11.42	13.5	4.2	307.4		
R.6965	Porphyritic basalt	80.07	82.95	8/8	16.4	−60.8	160/70	46.9	−1.5	114.3	5.5	7.5	324.5		
R.6966	Porphyritic basalt	80.07	82.95	12/13	211.7	14.3	165/63	204.1	−31.1	27.8	8.4	−7.7	300.3		
Mean in situ direction				16/16	44.0	−74.1				7.8	14.2				
Mean tilt-corrected direction								48.9	−1.4	19.4	8.6				
Mean pole												7.3	326.3	38.2	6.0

Note. k and α_{95} (K and A_{95}) are precision parameter and 95% confidence limits on direction (pole) following Fisher (1953).
[a] n is the number of samples included in the site mean and N is the total number demagnetized.
[b] Dec and Inc are declination and inclination of the paleomagnetic direction.
[c] Tilt corrections are given in dip to right of strike format.
[d] VGP is virtual geomagnetic pole given for a south pole.

cay observed in Type I samples (fig. 5*D*, 5*E*). The final type of demagnetization is shown in figure 5*F*. Sites R.6952, R.6964, and R.6965 have a higher NRM intensity, up to 5.0×10^{-2} Am^{-1}, carried by a mixture of magnetite and hematite with hematite dominant. Thin sections reveal that hematite is much more pervasive in these samples, a consequence of their having a different original composition with more ferromagnesian minerals. Thermal treatment to 580°C removes up to 10% of the NRM; demagnetization above this temperature causes the remanence to decay rapidly as a single component of magnetization toward the origin at 680°C. These high-temperature endpoints are taken to be the ChRM for samples at these sites and are used to calculate the mean direction (fig. 6*F*).

Although the demagnetization style and the magnetic carriers vary between the sites from the volcanic rocks, they record a consistent ChRM, directed to the northeast or southwest with a shallow inclination after tilt correction. Except for site R.6964, the magnetic polarity of samples within individual sites is consistent, but normal and reverse polarities are recorded at different sites (table 2). Site R.6964 has samples with both polarities within the site. However, because of the limited availability of rock exposure that could be drilled easily, samples for this site were collected from two distinct flows separated by a flow breccia. The polarity is consistent within the individual flows; but to obtain a site mean direction, some of the sample directions are inverted (fig. 6*F*).

Constraints on the Age of Magnetization. As stated earlier, the sedimentary and volcanic rocks record site mean directions with opposing magnetic polarities between sites. There is no apparent relationship between polarity and lithology, sampling locality, or magnetic carriers. Using the reversal test of McFadden and McElhinny (1990), the angle between the in situ polarity groups is 21°, yielding an indeterminate test at the 95% confidence level. Conversely, the tilt-corrected polarity groups are separated by only 4°, giving a positive test with a C classification. This suggests that reversals observed in the tilt-corrected data are real, while those seen in situ are an artifact due to the steep dip of the strata. The positive reversal test on the tilt-corrected data therefore supports a prefolding origin of the magnetization and indicates that it was acquired over a substantial time period.

Samples were collected from sites in both limbs of the Liberty Hills anticline and its minor parasitic folds to allow a fold test. There is, however, a problem with applying a standard fold test, since there are different numbers of apparently normal and reversed polarity sites in tilt-corrected and in situ coordinates (table 2; fig. 7). The test must be applied to site mean directions of the same polarity to determine whether the dispersion is greater in tilt-corrected or in situ coordinates. The usual approach is to invert directions from one polarity group in situ and then apply the test. If the interlimb angle of the fold is relatively large, the test works satisfactorily. However, in our case the interlimb angle is close to 0°, and site mean directions have to be rotated very large amounts to move them from in situ to tilt-corrected positions. Consequently, some sites with similar in situ directions, but opposite directions of tilt correction, will result in tilt-corrected site directions with approximately opposite polarities. One must therefore adopt an initial hypothesis that the remanence is either pre- or post-folding, so that the appropriate site mean directions can be inverted for the test. If we adopt the hypothesis that the remanence is prefolding and invert the in situ directions of sites R.6951–R.6955, R.6957, R.6959, R.6960, R.6962, R.6963, and R.6966 (table 2), there is an improvement in grouping with increased unfolding (fig. 8*A*). When the precision parameters (k) of the in situ and fully tilt-corrected directions are compared, the fold test is positive ($k_1/k_2 = 12.9$, >95% confidence limit of 1.88; McElhinny 1964). However, if we adopt the opposite hypothesis, that the remanence is post-folding, the in situ directions of sites R.6951, R.6956, R.6957, R.6960, R.6962, R.6963, and R.6966 are inverted (table 2). The grouping decreases with increased unfolding (fig. 8*A*), and comparison of the in situ and fully tilt corrected precision parameters demonstrates that the test is negative ($k_1/k_2 = 0.15$, <95% confidence limit of 1.88; McElhinny 1964). The data can thus be used to support either hypothesis. However, we have additional important information to discriminate between these hypotheses. After inversion of the appropriate site mean directions, the grouping achieved after fully tilt correcting the site mean directions is better than the best possible in situ grouping (fig. 8*A*). Comparison of the precision parameters demonstrates that the site mean directions are best grouped in tilt-corrected coordinates, and the improvement is statistically significant ($k_1/k_2 = 2.49$, >99% confidence limit of 2.44; McElhinny 1964).

Another problem is that the remanence may have been acquired during deformation, and the difficulties with applying the fold test may be a consequence of the remanence being synfolding in origin. It seems unlikely that a synfolding magnetization could lead to site mean directions

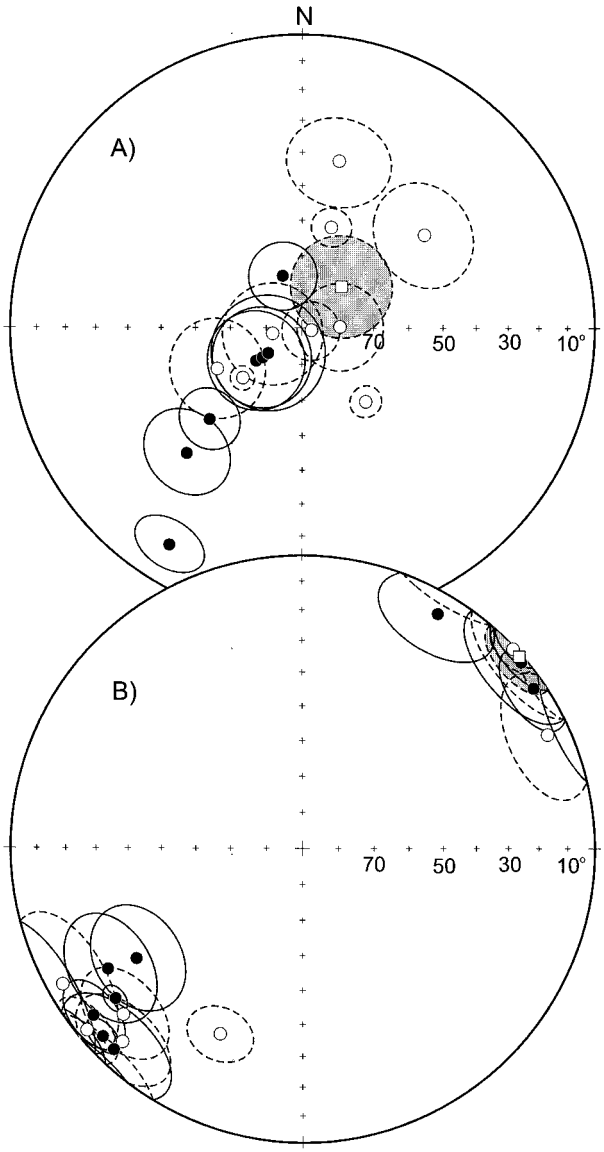


Figure 7. Equal-area stereoplot projections of site mean directions with their 95% confidence limits in *A,* in situ, and *B,* tilt-corrected coordinates. Open symbols denote upper-hemisphere plots; solid symbols denote lower-hemisphere plots. The formation mean direction, after conversion of all data to the upper hemisphere in *A* and the NE quadrant in *B,* is shown by a square symbol, and its associated 95% confidence limit is shaded.

with both polarities in different rocks with a range of magnetic carriers. However, to test for a possible synfolding remanence, an incremental fold test was again used, but site mean directions were inverted at each unfolding step as necessary to achieve the maximum precision parameter for each unfolding increment. Unfolding the data produces a series of k values, and the maximum k is achieved at 100% unfolding, indicating that the remanence is not a synfolding remanence (fig. 8*B*). Together, the data from the fold tests indicate that the best grouping of site mean directions is achieved after completely unfolding the tectonics and signify that the remanence predates the onset of the Gondwanan deformation in the Liberty Hills.

The new data presented here are currently the best constrained from the EWM in terms of the age controls, number of sites, range of sampled lithologies, and field tests. Nevertheless, while the fold test constrains the remanence as being older than Permo-Triassic and the reversal test indicates that the remanence was acquired over a long time pe-

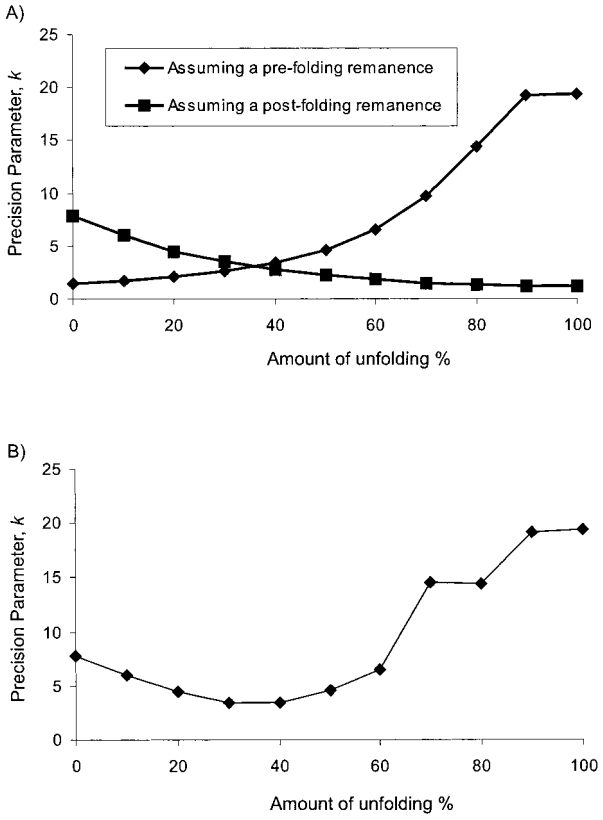


Figure 8. Effect of progressively unfolding the tectonics in 10% incremental steps on the precision parameter (k) of the formation mean direction. *A,* The effect of assuming pre- and postfolding remanences and applying the fold test in the usual way. The data can be used to support either hypothesis. However, the best grouping of tilt-corrected directions is statistically better than the best grouping of in situ site means. *B,* At each step, site mean directions were inverted as necessary to achieve the maximum possible k for that particular distribution of directions to identify a possible synfolding magnetization. The best grouping of data was achieved at 100% unfolding, and the improvement in k between the in situ and tilt-corrected mean is sufficient to pass a fold test at the 99% confidence level.

riod, these are not conclusive proof that the magnetization is primary. However, the range of magnetic carriers for the ChRM would suggest that the rocks of the Liberty Hills Formation have not been remagnetized. Although possible, it seems unlikely that the remanence results from a chemical remagnetization, since there is no evidence of a secondary magnetic mineral phase in the sedimentary rocks or of alteration of the volcanic rocks after the deuteric alteration. Furthermore, the remagnetization would have to have reset the magnetite and hematite over a long enough period that sites with both polarities are associated with both phases. Petrologic analysis indicates that the rocks of the Ellsworth Mountains have only been heated to temperatures experienced during subgreenschist facies metamorphism (Bauer 1992). These are not high enough to reset the magnetization of magnetite or hematite, and it is unlikely that the remanence is a thermoviscous remagnetization. Finally, the equatorial paleolatitude of the pole limits it to being of early Paleozoic age, since after the Early Silurian the pole for Gondwana moves progressively toward higher latitudes (Van der Voo 1993; Grunow 1999). As discussed earlier, there is no evidence for a tectonothermal event in the Ellsworth Mountains during the early Paleozoic that could reset the magnetization, and from the Early-Middle Ordovician onward, the EWM formed part of a passive margin. The interpretation preferred here is that the magnetization carried by the Liberty Hills Formation is a primary late Middle–early Late Cambrian remanence. A paleomagnetic pole for the Liberty Hills Formation has therefore been calculated, based on the tilt-corrected data, and a combination of virtual geomagnetic poles from all 16 sites yields a paleopole at lat 326.3°E and long 7.3°N, A_{95} = 6.0°.

Discussion

Other Paleomagnetic Poles from the EWM. The position of the EWM crustal block in early Paleozoic Gondwana reconstructions has previously been constrained by two paleomagnetic poles. The statistically indistinguishable poles are from the early Late Cambrian Frazier Ridge Formation (lat 4°N and long 296°E, A_{95} = 9°; Watts and Bramall 1981) and undated red calcareous siltstones at Nash Hills (lat 7°N and long 292°E, dp, dm = 4.2°, 8.4°; Grunow et al. 1987). The new Liberty Hills pole is clearly different from these earlier studies, lying over 30° of an arc away. The reason for this is uncertain. As the discrepancy between the poles is dominantly one of longitude, the possible influence of vertical-axis rotation must be considered. Such tectonic rotations are generally associated with regions that have experienced significant distributed strike-slip or transpressional deformation, with vertical-axis rotation facilitated by the development of cross faults, suborthogonal to the master strike-slip fault or plate margin. Although Curtis (1998*b*) has attributed the deformation of the Ellsworth Mountains to dextral transpressional tectonics, the dextral component of shear has been partitioned into discrete shear domains, while most of the range experienced distributed coaxial deformation associated with the pure-shear component. Late-stage,

mesoscale brittle faults, formed at a high angle to the regional structural grain and displaying a dextral displacement sense, have been described (Curtis 1998*a*), but they exhibit only centimeter-scale displacements and appear to form part of a conjugate set associated with postorogenic extensional collapse. The possibility that larger-scale cross faults are present and unexposed beneath transverse glacier systems cannot, however, be totally discounted. For such hypothetical faults to have facilitated vertical-axis rotations of discrete blocks, such that the paleomagnetic poles derived from Frazier Ridge/Nash Hills and Liberty Hills differ, would require one block to have rotated differentially from the other. This can be confidently discounted since the structural trend throughout the Heritage Range is constant, with strike varying by <5° between Frazier Ridge and Liberty Hills (Curtis 1998*b*).

Another possibility is apparent polar wander (APW), resulting from age differences between the formations. While the Liberty Hills pole is suggested to be late Middle to early Late Cambrian, the Frazier Ridge pole is dated as early Late Cambrian (Shergold and Webers 1992). Even taking the extremes of the ages and the paleomagnetic error limits requires a high-APW rate of 22 cm yr^{-1}; this explanation therefore seems unlikely.

A final possibility is that either the Liberty Hills or the Frazier Ridge/Nash Hills rocks record a remagnetization. Positive fold tests in the Frazier Ridge and in this study indicate that any remagnetization is pre-Permo-Triassic. Furthermore, the paleolatitudes indicate that both magnetizations must date from the early Paleozoic. The available constraints lead us to prefer the interpretation that the Liberty Hills Formation carries a primary remanence. The similarity of the Nash Hills and Frazier Ridge poles is generally interpreted as supporting the integrity of the data sets. Not only do they have similar pole positions but they also have similar magnetic properties—in both cases the remanence is held solely by a hematite carrier, and both record all sites with the same polarity. However, the Nash Hills pole is based on only 12 samples from partially remagnetized red calcareous siltstones of uncertain stratigraphic position. No correlative red calcareous sedimentary rocks were observed in the EWM during this study, and the age of the Nash Hills samples is therefore unknown. It is possible, therefore, that there could be a time gap between the deposition of the Frazier Ridge Formation and the Nash Hills siltstones, and the similarity of the magnetizations may indicate a regional early Paleozoic remagnetization event. If a remagnetization is the cause of the discrepancy, the reason why the overprint is not recorded in the Liberty Hills rocks is unclear. In addition, there is no evidence of an early Paleozoic tectonic event that could remagnetize the rocks. Therefore, without a comparison with new Frazier Ridge Formation data, the remagnetization hypothesis is impossible to substantiate. Hence, the source of the inconsistency in the early Paleozoic EWM poles remains uncertain.

Gondwanan Paleomagnetic Poles. The late Early to early Late Cambrian (from here on referred to as mid-Cambrian) paleomagnetic data for East Gondwana cratonic blocks are listed in table 3, with all available data from Australia and East Antarctica. To allow some impression of the quality of the poles, they are graded according to the reliability classification of Van der Voo (1990, 1993). In this classification, each pole is given a quality (Q) rating according to how many of seven criteria the pole fulfills. We regard the Q rating primarily as an information criterion. A high-Q rating does not necessarily prove that the paleopole is an accurate representation of the ancient geomagnetic field; it merely increases our confidence that this is the case. Only poles having $Q \geq 3$ are listed in table 3.

Several previous compilations of early Paleozoic Gondwanan poles have been made recently (e.g., Li and Powell 1993; Grunow 1995). Poles in table 3 are broadly similar to these previous studies, but the absolute ages assigned to sedimentary units have changed substantially because of modifications of the Cambrian time scale.

Three poles of possible mid-Cambrian age are presently available from East Antarctica, all with a Q rating of 3. There are, however, no reliable isotopic ages for the Mawson Station charnokite (Funaki and Saito 1992) and Ongul Island gneiss (Funaki and Wasilewski 1986) poles (Grunow 1995). Consequently, they cannot be confidently used for any tectonic interpretations and are excluded from further analysis. The remaining pole is from the Sør Rondane intrusive rocks (Zijderveld 1968). Although previously regarded as Ordovician in age, Grunow (1995) makes a strong case for the rocks having acquired their magnetization in the Early Cambrian, based on comparisons of isotopic closing and magnetization acquisition temperatures. The pole position also lies closer to the Cambrian rather than the Ordovician paleomagnetic data, and the 515 Ma age suggested by Grunow (1995) is accepted here.

Another eight poles are available from the Australian paleomagnetic database (table 3). The Q = 3–rated pole from the Hudson Formation (Luck 1972) lies some 45° from the mean of the rest of

Table 3. Gondwana Paleomagnetic Poles of Late Early to Early Late Cambrian Age in an Antarctic Reference Frame

Pole	Formation/sample section	Stratigraphic age/ isotope age range	Assigned age (Ma)	Latitude °N	Longitude °E	K	A_{95} (°)	Q
Antarctica:								
1	Mawson Station charnockite[a]	~966–426 Ma	?	−9	022	…	…	3
2	Ongul Island, gneiss[b]	~460–~530 Ma	500?	−17	010	…	…	3
3	Sør Rondane intrusives[c]	…	~515[d]	−28	010	…	5	3
Australia:								
4	NW Tasmanian Sediments[e]	Early Upper Cambrian	498	−11	022	20	10	7
5	Pantapinna Frome[f]	Late Middle Cambrian	502	−26	012	32	17	3
6	Areyonga Section[f]	Middle Cambrian	505	−25	016	574	5	6
7	Lower Lake Frome[g]	Middle Cambrian	505	0	024	10	15	3
8	Basal Lake Frome[f]	Middle Cambrian	505	−18	014	15	13	5
9	Hudson Frome[h]	Middle Cambrian	505	25	033	7	14	3
10	Billy Creek/Aroona Creek[f]	Late Lower–early Middle Cambrian	510	−22	006	11	14	6
11	Kangaroo Island sediments[f]	Late Lower Cambrian	515	−18	003	9	12	3

Note. Latitude/longitude are the positions of the paleomagnetic pole. K and A_{95} are Fisher (1953) statistical parameters. Q is the pole's quality rating (Van der Voo 1990, 1993). Poles 1, 2, and 9 are excluded from the compilation (see text for explanation). Rotation pole for Australia to Antarctica: long 038.9°E and lat 2.0°S, Δ = −31.5° (Royer and Sandwell 1989).
[a] Funaki and Saito 1992.
[b] Funaki and Wasilewski 1986.
[c] Zijderveld 1968.
[d] Age assignment taken from Grunow (1995).
[e] Li et al. 1997.
[f] Klootwijk 1980.
[g] Embleton and Giddings 1974.
[h] Luck 1972.

the data. Luck (1972) reported problems with recovering a ChRM from the samples, due primarily to a present geomagnetic field-directed chemical remagnetization and was uncertain whether the direction recovered was primary. Given the difference in the position of the pole when compared with the rest of the data and the problems reported with the demagnetization, the validity of the Hudson Formation result is in some doubt; therefore, it is excluded from the remaining analysis.

The mid-Cambrian poles in table 3 (other than excluded poles 1, 2, and 9) are shown in figure 9. There is some scatter, but this is not obviously age controlled and, therefore, suggests that it is not a consequence of APW. The poles, however, are inseparable (within error), suggesting that Gondwana was quasi-static during this period. One pole (7) from the Lower Lake Frome rocks (Embleton and Giddings 1974) lies away from the main group, but by an insufficient amount to warrant its exclusion. However, the significance of this pole is uncertain. It is also noteworthy that the 95% confidence limits on several of the poles are large, and further work is still needed to validate the Gondwana pole database. For comparison with the data presented in this article, a mean East Gondwana pole has been calculated from the available late Middle to early Late Cambrian data (505–498 Ma; table 4), the likely age of the Liberty Hills Formation.

Tectonic Implications

Comparison with Gondwana Poles. To test whether the new paleomagnetic data are consistent with the late Middle–early Late Cambrian location of the EWM suggested by the geologic information, three possible reconstruction models are considered. Models 1 and 2 are based solely on the Liberty Hills paleomagnetic data and are made by rotating the Liberty Hills pole to the mean of the Gondwana reference pole data and applying the same rotation to the EWM. The position of the Antarctic part of Gondwana in the early Paleozoic means that uncertainty exists as to what constitutes normal and reversed polarity. For model 1 (fig. 10), it is assumed that the calculated pole position of long 326.3°E and lat 7.3°N (table 2) is a south pole. For model 2, it is assumed to be a north pole. In both scenarios, tectonic motion of the EWM to their present-day position with respect to East Antarctica requires a rotation of ~90°, but in opposite senses (fig. 10). In addition, the reconstructions require that the EWM undergo a latitudinal translation. The early Paleozoic locations suggested in models 1 and 2 are

Table 4. Mean Gondwana Pole

Time period	No. of studies	Latitude °N	Longitude °E	K	A_{95} (°)	Poles included
505–498 Ma	5	−16	018	47	11	4, 5, 6, 7, 8

Note. Latitude/longitude are the positions of the paleomagnetic pole. K and A_{95} are Fisher (1953) statistical parameters. Poles 1, 2, and 9 are excluded from the compilation (see text for explanation). Rotation pole for Australia to Antarctica: long 038.9°E and lat 2.0°S, Δ = −31.5° (Royer and Sandwell 1989).

latitudinally displaced (by approximately 30°) from the geologically constrained position, which places the EWM in or close to the Natal Embayment (fig. 1*B*) and the other Gondwanan continents (fig. 10). When the rotation required to map the Liberty Hills pole onto the Gondwana mean pole is applied to the Frazier Ridge and Nash Hills poles, both move toward the Gondwana data, but by an insufficient amount to result in an overlap of their respective confidence limits. These EWM data are therefore inconsistent with both models. If model 1 or 2 were to represent the correct paleogeographical position of the EWM in the late Middle to early Late Cambrian, moving the block to its present position requires an approximate 90° rotation and substantial dextral translation of the EWM along the East Antarctic margin. This is broadly similar to the model of Schmidt and Rowley (1986), who placed the EWM in a prebreakup position off the East Antarctic coast near the Queen Maud Mountains.

Intrinsic problems exist, however, with both models 1 and 2, since they suggest that the EWM originated from a similar paleolatitude to northern Victoria Land, which formed part of the Cambro-Ordovician active margin, recorded in the Transantarctic Mountains and eastern Australia as the Ross and Delarmerian orogenies, respectively. There is little supporting evidence for such a deformation event having occurred in the EWM. To the contrary, an excellent Middle-Late Cambrian correlation exists between the rifting events in the EWM and South Africa (Curtis et al. 1999). In addition, isotopic evidence from igneous rocks within the EWM block indicates that the mountains are underlain by Grenvillian-age continental crust (Storey et al. 1994; Curtis et al. 1999), supporting their position near the Grenville front in Coats Land (fig. 1*B*). Furthermore, assuming that the rotation of the EWM occurred as a single event during Gondwana breakup, a position suggested by models 1 and 2 results in a highly oblique relationship between the Permo-Triassic structural grain of the EWM and the East Antarctic paleo-Pacific margin (models 1 and 2, fig. 10). The EWM would also have to have lain "outboard" of the Paleozoic subducting margin of Gondwana, and motion to the present position would be encumbered by the West Antarctic continental margin terranes of Marie Byrd Land, Thurston Island, and the Antarctic Peninsula (fig. 1*A*). Given the geologic evidence and the foregoing discussion, the 30° latitudinal separation between the EWM and the Natal Embayment, predicted by the Liberty Hills paleomagnetic data, must be considered to be, at the very least, geologically unlikely.

Model 3 of figure 10 demonstrates the best-fit reconstruction based on the geologic evidence alone. Here the EWM are rotated so that they occupy a prebreakup position in the Natal Embayment, allowing the strong tectonostratigraphic correlations that exist between the EWM and southern Africa to be satisfied. After applying the same rotation to the paleomagnetic data, it is clear that the Liberty Hills pole is inconsistent with this reconstruction (fig. 10), being latitudinally displaced from the mean of the Gondwana data. When the EWM are restored to a Natal Embayment location, the Frazier Ridge and Nash Hills poles also lie away from the Gondwana mean, but their confidence envelopes do overlap with the confidence limit of the Lower Lake Frome pole (7), although it should be reiterated that the Lower Lake Frome pole is of uncertain value. Thus, these data might be considered broadly consistent with the geologically preferred reconstruction, but the fit is poor.

Preserving the amounts of rotation suggested by the Liberty Hills pole in models 1 and 2 (approximately 90°) and simply allowing a more southerly paleolatitude for the EWM permits alignment of the Permo-Triassic structural grain of the EWM and Cape Fold Belt, alleviating the unacceptable structural discordance inherent in a position close to northern Victoria Land. While the difference between the Liberty Hills and the Frazier Ridge and Nash Hills poles is essentially one of longitude, it is noteworthy that the discrepancy that exists between the mean Gondwana pole and all of the EWM data is essentially one of latitude.

If the Liberty Hills pole is an accurate record of the late Middle to early Late Cambrian geomagnetic field, and there is no evidence to suggest otherwise, the pole presents an enigma in Gondwana reconstructions. There exists a dichotomy between the paleomagnetic data that would place the EWM near northern Victoria Land and the overwhelming

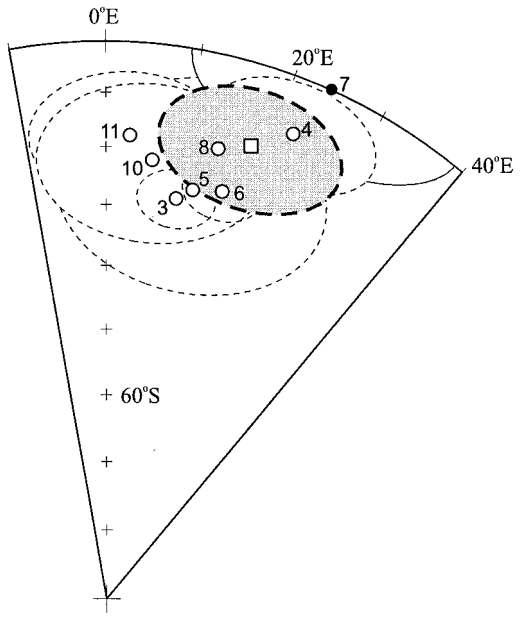


Figure 9. Equal-area stereoplot of late Early to early Late Cambrian poles from Gondwana (table 3), excluding poles 1, 2, and 9 (see text for explanation). Open symbols denote southern hemisphere plots; solid symbols denote northern hemisphere plots. The mean pole of the late Middle to early Late Cambrian data (505–498 Ma pole, table 4) is shown by a square, and its associated 95% confidence limit is shaded.

geologic evidence that would place the block close to southern Africa. Two other possible sources of error also require consideration. First, the plate-reconstruction parameters used to rotate the data and the major continents into an Antarctic reference frame are a possible source of error. The parameters of, for example, Martin and Hartnady (1986), Lawver and Scotese (1987), and Lottes and Rowley (1990) all produce different positional relationships between Africa, East Antarctica, and Australia. Since there are no well-dated poles of exactly the same age in the Antarctic, Australian, and African data sets, it is impossible to identify problems with the plate-reconstruction parameters. However, given the range of available reconstructions, a 30° latitudinal discrepancy between the EWM and Gondwana paleomagnetic data cannot be solely the consequence of errors in Gondwana reconstructions. Second, the Gondwana reference poles themselves are far from being well constrained. However, as far as can be ascertained, the consistency of the data suggests that these poles accurately represent the late Middle to early Late Cambrian geomagnetic field. It is noteworthy, though, that in several early Paleozoic apparent polar wander paths (e.g., Klootwijk 1980; Schmidt et al. 1990; Van der Voo 1993), the south pole remains close to North Africa from the earliest Cambrian to possibly the Silurian. Poles from a recent compilation of Paleozoic Gondwana data by Grunow (1999) are shown in figure 11, along with the late Middle to early Late Cambrian data used for reference in this study. It is clear that at the 95% confidence level, many of the presumed Cambrian data from Gondwana are also in accordance with the Early Ordovician to Early Silurian Gondwanan data, with three of the five supposed Cambrian poles lying within the confidence limit of the Early Silurian pole. While it is possible that the apparent polar wander path passed through the same position several times, the possibility that some of the presumed Cambrian Gondwana poles have been remagnetized cannot be completely excluded. Crucially, only one of the Cambrian poles (that of Li et al. 1997) has field tests that constrain the age of magnetization. Remagnetizations would be difficult to identify in cratonic Gondwana rocks of this age, and this may be a possible explanation for the discrepancy between the EWM and Gondwana data after making the preferred geologic reconstruction.

The reason for the inconsistency between the well-constrained paleomagnetic data and the multiple lines of evidence that suggest a Natal Embayment position for the EWM still remains uncertain at this time. Of greater significance, perhaps, is the presently unresolved difference between the Frazier Ridge, Nash Hills, and Liberty Hills paleomagnetic data from the Ellsworth Mountains themselves.

Conclusions

Paleomagnetic data from the Liberty Hills Formation define a new, well-constrained, late Middle to possibly early Late Cambrian pole for the EWM. Reconstructions using the pole suggest a geologically unlikely prebreakup position for the EWM near northern Victoria Land, some 30° of latitude away from the geologically preferred position in the Natal Embayment. The reasons for this remain uncertain, but problems with the plate-reconstruction parameters used to transfer the poles from an Australian to an Antarctic reference frame or, more likely, uncertainties in the Gondwana pole database may play a part. More well constrained paleomagnetic data are required from the EWM, including data from the Frazier Ridge Formation, to allow evaluation of the new results and explanations of the differences between the EWM Cambrian paleomagnetic data. To refine further the early Paleozoic

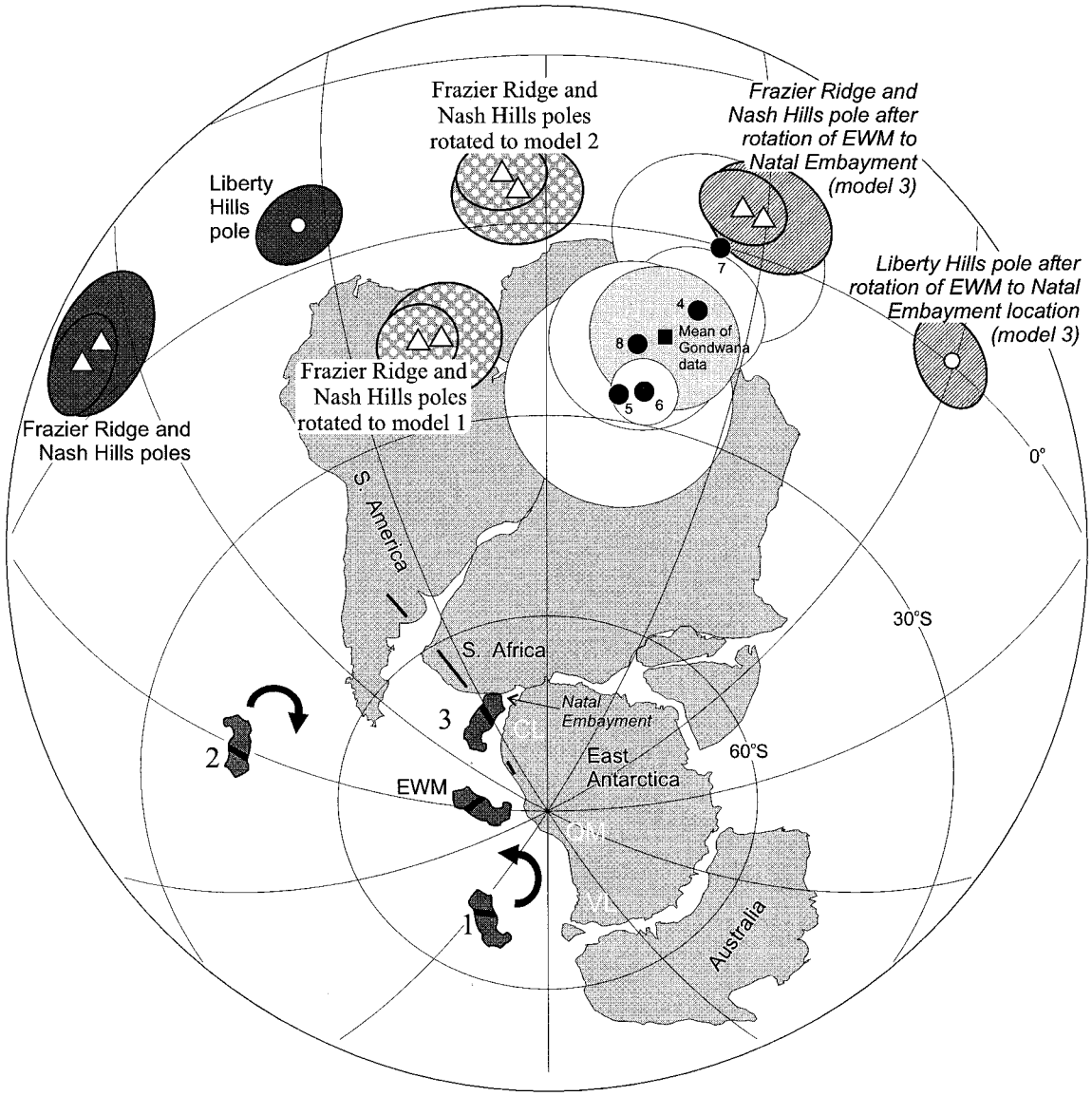


Figure 10. Possible paleogeographic reconstructions of the Ellsworth-Whitmore Mountains (EWM) within Gondwana and their effect on the available paleomagnetic data. Reconstruction is made by keeping East Antarctica fixed and rotating all continents into, and treating all poles in, an Antarctic reference frame. Fit of major continents is from Lawver and Scotese (1987). Black circles (numbered *4–8*) are the late Middle to early Late Cambrian reference poles (see table 3). For the EWM: model 1, fit using new Liberty Hills paleomagnetic pole, assuming that site mean directions in the NE quadrant are reversed polarity (cf. Watts and Bramall 1981); model 2, fit using Liberty Hills pole, assuming opposite polarity. These two positions require opposite senses of rotation of the EWM to translate the block into its present position. The effect of both of these fits on the Frazier Ridge (Watts and Bramall 1981) and Nash Hills (Grunow et al. 1987) data is also shown. Model 3 is the best fit, based on the geologic evidence, achieved by rotating the EWM about a Euler pole at long 337.0°E and lat 82.1°S, $\Delta = -93°$. *EWM* = present-day position of the EWM crustal block; *VL* = northern Victoria Land; *QM* = Queen Maud Mountains; *CL* = Coats Land. Black lines show orientation of Permo-Triassic Gondwanan fold belts: Cape Fold Belt in southern Africa, Sierra Australes fold belt in South America, Pensacola Mountains in East Antarctica.

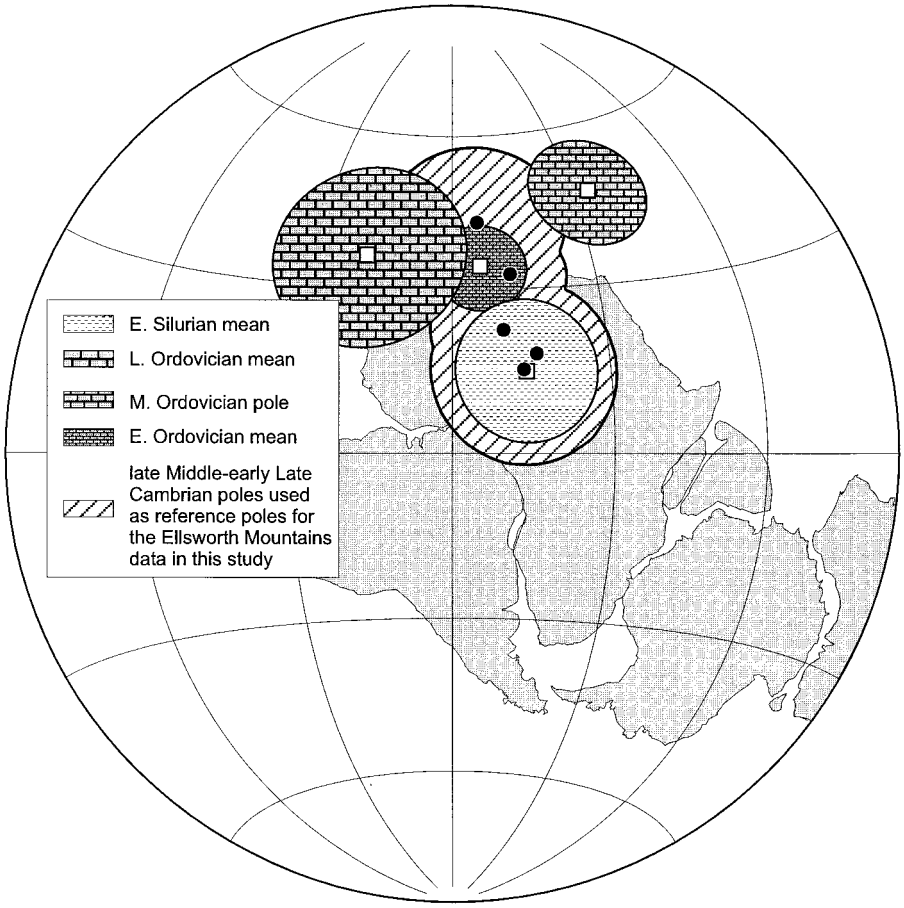


Figure 11. Late Middle Cambrian to Early Silurian Gondwanan paleomagnetic poles. Black circles are the late Middle to early Late Cambrian poles, used as a paleomagnetic reference for this article (table 3). White squares are the mean poles for the Early Ordovician to Early Silurian; in the case of the Middle Ordovician, the white square is the pole of the only available study (Grunow 1999). Error circles are shown for 95% confidence. Plot is made by keeping Africa fixed and rotating all data and continents into an African reference frame using the following rotation parameters: Australia to Antarctica, long 038.9°E and lat −2.0°N, Δ = −31.5°; India to Antarctica, long 10.9°E and lat 7.8°N, Δ = 65°. (Royer and Sandwell 1989); Antarctica to Africa, long 328.58°E and lat −7.78°N, Δ = −58.0° (Lawver and Scotese 1987).

position of the EWM block and the tectonic evolution of Antarctica, better-constrained early Paleozoic reference poles are required for the major Gondwana continental blocks.

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